Bachelor in Artificial Intelligence

AI for Communication and Marketing

LAB EXAM PART 3

Customer Lifetime Value & Marketing Strategy

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THE CUSTOMER BASE

WHO ARE WE TALKING ABOUT?

61,872
existing customers analysed

97%
one-time buyers

2.91%
repeat purchase rate

1,117
active customers

THE REVENUE PICTURE

PREDICTED vs ACTUAL HOLDOUT

BRL 80,072
BTYD predicted 90-day CLTV
existing customers only

63× gap

BRL 5,073,141
actual holdout revenue (5 months, all customers)

Because 98% of holdout revenue comes from new customers acquired after the calibration period — outside the scope of any BTYD or ML model.

MODEL RECOMMENDATION

FOR BUDGET FORECASTING

★ BTYD: BG/NBD + GAMMA-GAMMA

① Alive probability per customer
Every customer gets a score 0–1 showing how likely they are to still be active.

② No labelled holdout required
BTYD estimates future behaviour from purchase history alone.

③ Same accuracy as ML
On customers who actually returned, all three models score similarly. BTYD wins on interpretability, not on raw numbers.

Olist is an acquisition-driven marketplace. BTYD quantifies the value of the existing base and identifies who is still active — the foundation for any retention or win-back investment.

Top 10% CLTV segment (6,214 customers · avg alive prob 0.897 · avg CLTV BRL 4.03) → **Retention target**
Win-Back pool: 516 customers · avg alive prob **0.087** · avg historical order BRL **228** · top candidate avg order BRL 800+

#1 ACTION
HIGH VALUE · HIGH DROPOUT PROBABILITY

WIN-BACK CAMPAIGN

516
target customers

0.087
avg alive probability

BRL 228
avg historical order value

TIER 1 – TOP 20

Personalized Voucher

avg order BRL 800+

Single recovered purchase = 4–7× platform avg revenue. Direct high-value outreach.

TIER 2 – RECENCY < 90d

Reactivation Email

~230 customers

Triggered email referencing lat purchase category. Lower dormancy = lower barrier.

TIER 3 – RECENCY > 90d

High-Incentive Offer

monetary > BRL 200

Bundle or free shipping offer. Stronger nudge needed for longer dormancy.

#2 ACTION
KEEP HIGH-VALUE ACTIVE

RETENTION STRATEGY

5,494 retention customers

0.999 avg alive prob

BRL 17K total CLTV 90d

Recommended Actions

- **Loyalty incentive**: prob_alive > 0.95 → early access or discount on next order
- **60-day nudge**: triggered email before model assigns low alive probability
- **VIP tier**: CLTV > BRL 50 → dedicated exclusive offer